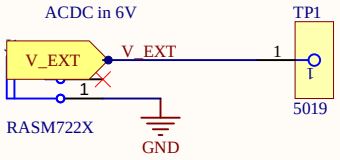
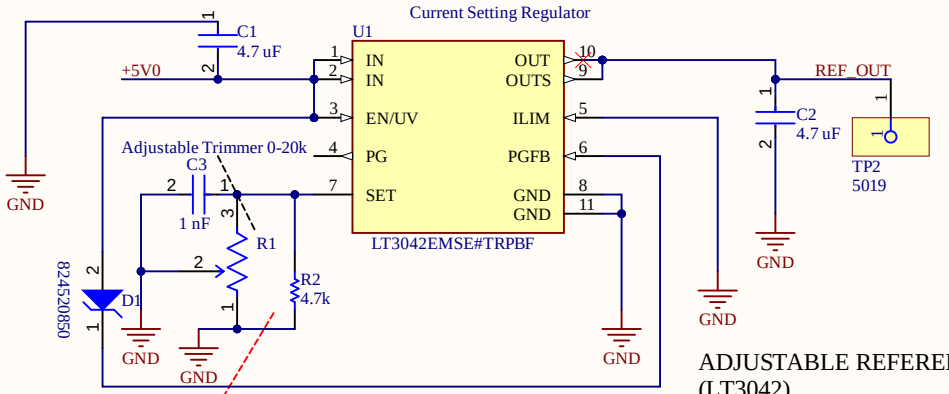
Power Regulation, PSRR Simulations

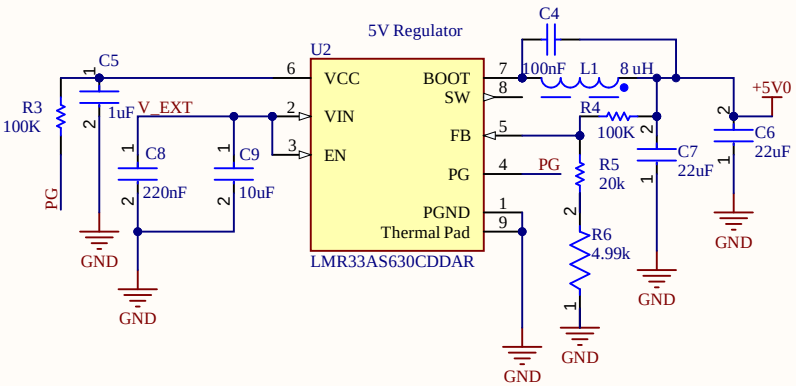


POWER INPUT (6 V DC)-
Input: External 6 V DC adapter-
Output: 6 V supply rail-
Function: Provides the primary DC power supply for the laser driver system.



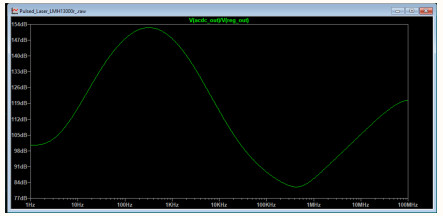
VREF = ISET × REQ
ISET = 100 µA
REQ = RPOT || 4.7 kΩ
RPOT = 0–20 kΩ
VREF = 0.1 × REQ (kΩ)
Adjustment Range: 0–0.381 V
Maximum Laser Current: ~190 mA

ADJUSTABLE REFERENCE REGULATOR (LT3042)
- Input: 5V DC
- Output: Adjustable low-noise reference voltage (VREF)
- Function: Generates a stable, adjustable reference voltage for laser diode current control.



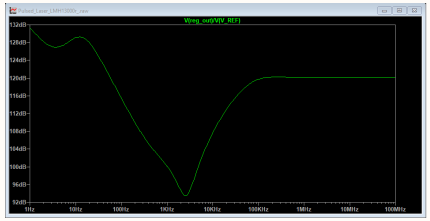
5V Regulator (LMR33A)
- Input: ACDC supply
- Output: Regulated 5V supply
- Function: Provides power to accessories and ICs

Note: PSRR values are simulated and may differ from datasheet specifications due to configuration and modelling assumptions.



5 V REGULATOR - PSRR ANALYSIS

AC sweep: 1 Hz–100 MHz
Input: 7 V DC, 1 V AC small-signal
Simulated PSRR @ 1 MHz: ~90 dB
Datasheet PSRR @ 1 MHz: 79 dB



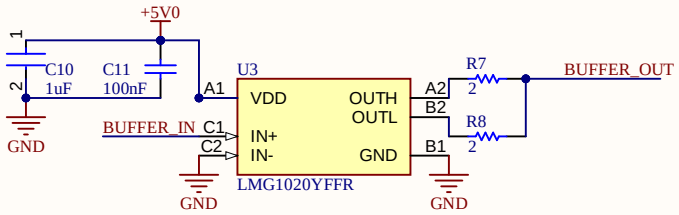
100 mV REFERENCE — PSRR ANALYSIS

AC sweep: 1 Hz–100 MHz
Input: 5 V DC
Simulated PSRR @ 1 MHz: ~121 dB
Datasheet PSRR @ 1 MHz: 79 dB (LT3042 nominal configuration)

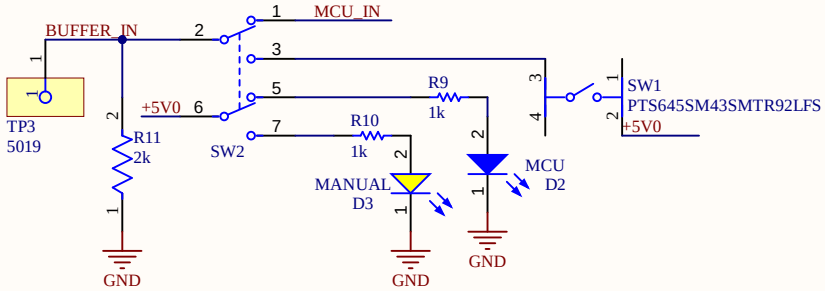
Program: **Quantum Key Distribution**	Drawn By: **Adam Szautner**	Reviewed By: *****	Revision: **1.0**
Project Title: **Laser Driver**	Project ID: **QKD-LD-001**	Last Modified: **Nov 1st, 2026**	SHEET 1 OF 3

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Pulse Generation, Indicator LEDS & Control

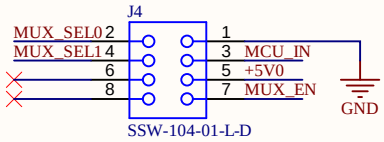


PULSE BUFFER (LMG1020)
- Configuration: High-speed, non-inverting pulse driver
- Input: Electrical pulse from DS1023
- Output: High-current, fast-edge electrical pulse
- Function: Buffers incoming nanosecond pulses and provides sufficient drive strength for transmission over 50 Ω coaxial cable, preserving pulse timing and width.



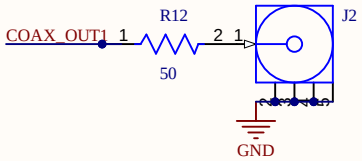
Program: **Quantum Key Distribution**	Drawn By: **Adam Szautner**	Reviewed By: **=RvwdBy**	Revision: **1.0**
Project Title: **Laser Driver**	Project ID: **QKD-LD-001**	Last Modified: **Nov 1st, 2026**	SHEET 2 OF 3

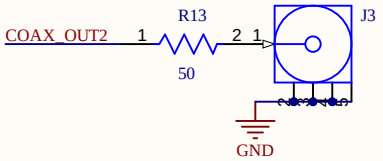
External Peripheral Interfaces



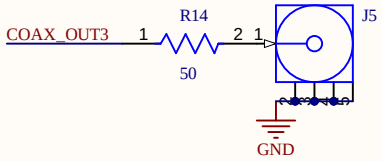
J4 - MCU Control

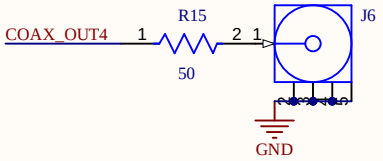
Supplies 5V and GND to MCU
Allows for switching of the 4 way switch
Input for MCU pulse triggering

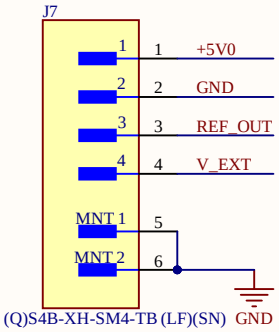




J3-J6 - SMA PULSE OUTPUT
High-Speed Pulse Control Outputs.
Termination at destination of coax







J7 - SECONDARY PCB POWER & CONTROL

Supplies 5 V, GND, and VREF to the secondary laser driver PCB. Fourth pin reserved for future use.

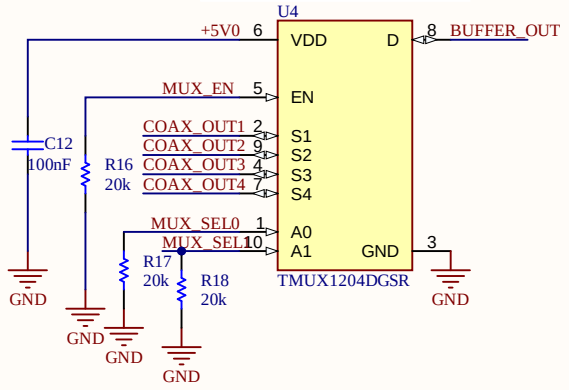
High-speed pulse signal supplied separately through J3 (SMA).

Continous current max of 30 mA, we shouldn't need to exceed this continously but worth keeping it in mind

U4 – HIGH-SPEED PULSE SELECTION

TMUX1204 4:1 analog switch routes MUX_IN to one of four coaxial outputs (COAX_OUT1–4).

Output selection controlled by MCU signals MUX_SEL0 and MUX_SEL1. MUX_EN provides independent enable control.



Program: **Quantum Key Distribution**	Drawn By: **Adam Szautner**	Reviewed By: **=RvwdBy**	Revision: **1.0**
Project Title: **Laser Driver**	Project ID: **QKD-LD-001**	Last Modified: **Nov 1st, 2026**	SHEET 3 OF 3

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Board Stack Report